

Vaibhav Singh

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SUMMARY

Computer Science Engineering student focused on hands-on systems programming projects, low-level emulators, backend services, and full-stack applications using Java, Spring Boot, and React. Strong interest in runtime environments, performance optimization, and scalable RESTful APIs.

EDUCATION

SRM UNIVERSITY, RAMAPURAM

B.Tech in Computer Science and Engineering

Coursework: Operating Systems, Computer Architecture, DBMS, Computer Networks, DSA
Chennai, India

June 2023 – Present

GPA: 9.2/10.0

ST. PAUL'S COLLEGE

12th ISC Board – 78%

10th ICSE Board – 90%

April 2021 – May 2022

April 2019 – May 2020

EXPERIENCE

Technical Team Member | INTELLECTS SRMIST Ramapuram

April 2024 – Present

Chennai, India

- Collaborated in a cross-functional team to architect 3 modular backend microservices using Java and Spring Boot, optimizing database indexing to reduce API response overhead by 25%.
- Spearheaded a campus competitive programming initiative, growing an active community to 100+ students.
- Conducted technical software development seminars for online campus events reaching 65+ attendees.

PROJECTS

NES Emulator | Java, Assembly, MOS 6502 Architecture

[mrborn2suffer/NES-or-Mess](https://github.com/mrborn2suffer/NES-or-Mess)

- Engineered a cycle-accurate MOS 6502 CPU emulator handling 56 base instructions over a 256-opcode matrix at 1.79 MHz.
- Developed a Picture Processing Unit (PPU) executing stable 60 FPS scanline rendering with sub-pixel scrolling.
- Integrated memory bank switching (NROM, MMC1, MMC3, AxROM) ensuring compatibility with 500+ commercial ROMs.

Aurora | Spring Boot, React, JWT, Relational DB

[mrborn2suffer/Aurora](https://github.com/mrborn2suffer/Aurora)

- Created a free digital reading application using Spring Boot and React featuring secure token-based JWT authentication.
- Designed a responsive layout optimized with content carousels and multi-criteria dynamic sorting features.
- Implemented 15+ RESTful API endpoints coupled with a relational database layer to handle user profiling and book discovery loops.

3D Raycasting Engine | Java, DDA Algorithm, Vector Math

[mrborn2suffer/Wolfstein](https://github.com/mrborn2suffer/Wolfstein)

- Utilizing the Digital Differential Analyzer (DDA) algorithm, coded a 3D graphics rendering engine from scratch for real-time grid ray projections.
- Applied linear algebra and vector mathematics to compute ray-wall intersections, resolving 100% of camera lens distortion across a 60-degree Field of View.
- Optimized the core execution loop, maintaining 60+ FPS using custom delta-time driven collision logic.

SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Frameworks & Libraries: Spring Boot, Quarkus, React.js, LibGDX, Selenium

Tools & Databases: Git, GitHub, Postman, Linux, Firebase, MongoDB, PostgreSQL, Maven, Docker

Core Competencies: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), RESTful APIs, Unit Testing, Operating Systems

CERTIFICATIONS

Oracle Certification for DevOps

[Link](#) · Oct 2025

AWS Training Certification for Machine Learning Basics

[Link](#) · Aug 2025

NPTEL Certification for Data Structures and Algorithms in Java

[Link](#) · Oct 2024

Smart India Hackathon (SIH 2025) Internal Nominee

[Link](#) · Sept 2025